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Effect of minimally invasive mitral valve surgery compared to sternotomy on short- and long-term outcomes: a retrospective multicentre interventional cohort study based on Netherlands Heart Registration

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Abstract

Key question

Are short- and long-term outcomes of patients undergoing MIMVS versus sternotomy comparable, based on real-world data?

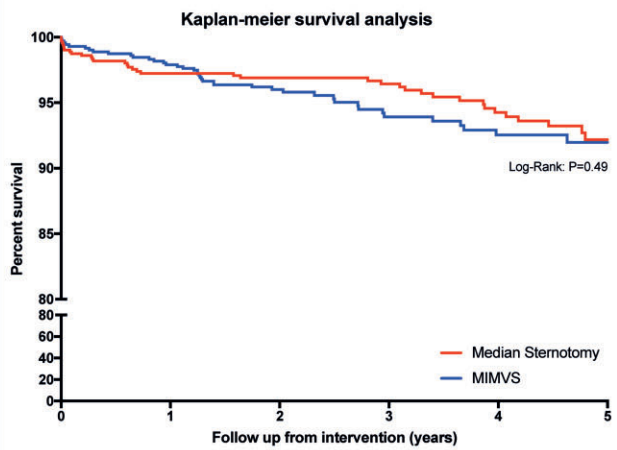
Key finding(s)

There are no differences in 5-year survival. MIMVS results in a lower repair rate and more reinterventions after 5 years.

Take-home message

MIMVS is as safe as sternotomy. In the real-world, MIMVS comes at a cost of a reduced repair rate and more reinterventions in the long term.

5-year Kaplan-Meier survival analysis for the matched population.



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OBJECTIVES: Minimally invasive mitral valve surgery (MIMVS) has been performed increasingly for the past 2 decades; however, large comparative studies on short- and long-term outcomes have been lacking. This study aims to compare short- and long-term outcomes of patients undergoing MIMVS versus median sternotomy (MST) based on real-world data, extracted from the Netherlands Heart Registration.

METHODS: Patients undergoing mitral valve surgery, with or without tricuspid valve, atrial septal closure and/or rhythm surgery between 2013 and 2018 were included. Primary outcomes were short-term morbidity and mortality and long-term survival. Propensity score matching analyses were performed.

RESULTS: In total, 2501 patients were included, 1776 were operated through MST and 725 using an MIMVS approach. After propensity matching, no significant differences in baseline characteristics persisted. There were no between-group differences in 30-day mortality (1.1% vs 0.7%, $P = 0.58$), 1-year mortality (2.6% vs 2.1%, $P = 0.60$) or perioperative stroke rate (1.1% vs 0.6%, $P = 0.25$) between MST and MIMVS, respectively. An increased rate of postoperative arrhythmia was observed in the MST group (31.3% vs 22.4%, $P < 0.001$). A higher repair rate was found in the MST group (80.9% vs 76.3%, $P = 0.04$). No difference in 5-year survival was found between the matched groups (95.0% vs 94.3%, $P = 0.49$). Freedom from mitral reintervention was 97.9% for MST and 96.8% in the MIMVS group ($P = 0.01$), without a difference in reintervention-free survival ($P = 0.30$).

CONCLUSIONS: The MIMVS approach is as safe as the sternotomy approach for the surgical treatment of mitral valve disease. However, it comes at a cost of a reduced repair rate and more reinterventions in the long term, in the real-world.

Keywords: Mitral valve surgery • Minimally invasive mitral valve surgery • Mitral valve repair • Nationwide registry

ABBREVIATIONS

CPB	Cardiopulmonary bypass
EuroSCORE I	European System for Cardiac Operative Risk Evaluation
LOHS	Length of hospital stay
MIMVS	Minimally invasive mitral valve surgery
MST	Median sternotomy
MVr	Mitral valve repair
MVR	Mitral valve replacement
NHR	Netherlands Heart Registration

INTRODUCTION

Minimally invasive mitral valve surgery (MIMVS) has an established role in the surgical treatment of mitral valve (MV) disease, but it is still not widely adopted. Over the past decades, many high-volume centres have published excellent outcomes, comparing MIMVS to sternotomy [1, 2]. These studies, and subsequent meta-analyses, demonstrated no differences in safety, defined as mortality, and efficacy, defined as the recurrence of mitral regurgitation (MR), despite longer cardiopulmonary bypass (CPB) and aortic cross-clamp (X)-times. Furthermore, a tendency towards reduced morbidity, with a lower incidence of re-exploration for bleeding, arrhythmia, pneumonia and wound infection, in conjunction with a shorter hospital stay and reduced postoperative pain has been observed [3, 4]. In addition, MIMVS has been associated with improved cost-effectiveness [5] and reduced mortality rates in a reoperative setting [6].

Long-term outcome of MIMVS has only been reported by expert single centres [2, 7–9] and 1 multicentre British group [10], while most multicentre studies have focused on short-term outcomes [11], resulting in a knowledge gap of long-term real-world data. Therefore, it remains debateable whether the published long-term outcomes are reproducible in real world. As such, the most recent European Society of Cardiology (ESC)/European Association for Cardiothoracic Surgery (EACTS) guidelines for valvular heart disease do not contain a section on the preferred surgical approach for the treatment of MV disease [12]. However, a consensus statement by The International Society of Minimally Invasive Cardiothoracic Surgery has been published a decade ago, in which some recommendations have been formulated, carried by a low level of evidence [13–15]. Subsequently, many centres remain apprehensive about the development of a minimally invasive programme, due to a presumed steep learning curve coupled with this lack of long-term evidence [16]. Furthermore, as the surgical outcomes are associated with the hospital and surgeon volume [17–19], there is a need for a case load that ensures sufficient volume and continuity. The aim of the current study was to evaluate the short-term and long-term outcomes of patients undergoing MV surgery in the Netherlands, registered in the Netherlands Heart Registration, either through a minimally invasive approach or sternotomy, to provide real-world evidence.

PATIENTS AND METHODS

Source of study data

Data from the national database of the Netherlands Heart Registration (NHR) were used for this study. All patients undergoing cardiac surgery in the Netherlands are collected in this prospective mandatory database. The basis for data collection was reported elsewhere [20]. In short, this prospective database contains a variety of data on cardiac surgical procedures performed in the Netherlands. In addition to demographic factors and type of intervention, the database includes parameters concerning perioperative morbidity and mortality and long-term survival and all risk factors required for the calculation of contemporary risk scores. Participation in this registry was made compulsory by national society decree. Although this database was initiated for the quality evaluation of cardiovascular interventions, its completeness of mandatory preoperative variables

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enables anonymized long-term follow-up of patients undergoing cardiac surgical interventions. All data were automatically uploaded and anonymized.

Inclusion

Patients were considered for inclusion if they were 18 years and older and underwent isolated primary MV surgery. Isolated MV surgery was defined as mitral valve repair (MVr) or replacement (MVR) combined with tricuspid valve repair/replacement and/or rhythm surgery (pulmonary vein isolation/MAZE procedure) and/or atrial septal closure. Patients undergoing a reoperation or other concomitant interventions were excluded. As robotic surgery is a distinct area within the field of MIMVS, patients undergoing a robotic approach were excluded as well. Minimally invasive surgery was defined as right anterolateral thoracotomy with peripheral cannulation and avoiding sternotomy. No distinctions were made between the length of the incision, or the use of a video-assisted or direct-vision approach.

Study design

The current study is a retrospective multicentre cohort study of prospectively collected data from 16 cardiothoracic centres in the Netherlands. Data were collected for patients undergoing MV surgery between 2013 and 2018, enabling complete long-term follow-up.

Outcomes

All preoperative data and baseline characteristics required for the calculation of the European System for Cardiac Operative Risk Evaluation (EuroSCORE I) were retrieved from the NHR database. Furthermore, short-term outcomes, defined as early mortality (either 30 days or in-hospital), postoperative complications (i.e. peri-operative stroke, vascular complications, myocardial infarction, pneumonia, wound infection, reintubation/prolonged ventilation, re-admission to the intensive care, renal failure, gastrointestinal complications, re-exploration for bleeding and arrhythmia) and duration of hospital stay, were retrieved. Long-term outcomes were defined as long-term survival at 5 years and reintervention for the recurrence of MV disease at 5 years. Mortality data were derived from the municipal administration records and were almost complete for all patients. Mitral valve reintervention was retrieved by cross-referencing the registry's database. Both mortality and reintervention follow-up were completed through 1 December 2020.

Missing values

The national database of the Netherlands Heart Registration has an exceptionally high data completeness for baseline characteristics. Unfortunately, 2 non-mandatory variables had a relatively high proportion of missing data: CPB-time (33.1%) and X-time (32.7%). Given the higher levels of missing data for these variables, which clearly exceed >10% and most likely had a non-random distribution, it was inappropriate to apply a multiple imputation method [21].

Statistical analysis

Normality of the continuous variables was tested by visual inspection of the histograms and the Shapiro–Wilk test. Continuous data are presented as mean ± standard deviation (or as median with interquartile range in the presence of skewedness). Categorical data are expressed as frequencies and percentages and were compared using the χ^2 test. Fisher's exact test was used when the minimum expected cell-size assumption did not apply. Continuous variables were compared using the Mann–Whitney *U*-test. Covariates, used for the calculation of propensity scores, were identified at baseline analysis. Propensity score matching analyses were used to correct for potential confounders. Propensity scores were estimated using variables identified in a logistic model. Propensity scores were matched using nearest neighbour matching in a 1:1 ratio, replacement was not allowed, and calliper distance was set at 0.02, as advocated by experts in our field [22]. Baseline characteristics were presented both before and after propensity score matching. Standardized mean difference was used to compare the difference in means in units of the pooled standard deviation. A value of >0.10 was considered an index of residual imbalance. Kaplan–Meier survival curves were used to demonstrate long-term survival and freedom from reoperation. Between survival curve differences were assessed using the log-rank test. All reported *P*-values were two-sided and were considered statistically significant when $P < 0.05$. Statistical analyses of the data were performed using SPSS software (V26, IBM, Armonk, New York, USA) and R Statistics (the R Foundation, Vienna, Austria).

Ethical statement

This is a retrospective study with data of the Netherlands Heart Registration. Therefore, no institutional review board approval was necessary. This study is in line with the institution's ethical policies and standards.

RESULTS

Baseline characteristics

The NHR database contained 2501 patients undergoing MV surgery with a registered surgical approach between 2013 and 2018, of which 1776 were operated through a median sternotomy (MST) and 725 using an MIMVS approach. Patients selected for MIMVS had a significant lower body mass index (24.8 [22.9–27.1] vs 25.6 [23.2–28.0]; $P < 0.001$), a lower rate of diabetes (4.1% vs 8.3%, $P < 0.001$), chronic obstructive pulmonary disease (4.6% vs 10.5%, $P < 0.001$), extra-cardiac arteriopathy (1.2% vs 3.0%, $P = 0.01$) and active endocarditis (0.1% vs 1.7%, $P < 0.001$) compared to patients selected for MST. Patients operated through a minimally invasive approach had significantly better left ventricular ejection fraction and lower rate of severe pulmonary artery hypertension. This resulted in a lower median EuroSCORE I (2.99 [1.90–5.36] vs 3.61 [2.08–6.22]; $P < 0.001$). Baseline characteristics are summarized in Table 1.

Propensity score matching

To correct the differences in baseline characteristics, propensity score matching was performed. Parameters that differed at baseline, i.e. body mass index, diabetes mellitus, chronic obstructive pulmonary disease, extra-cardiac arteriopathy, active endocarditis, left ventricular ejection fraction, pulmonary artery hypertension, EuroSCORE I and concomitant procedures (tricuspid valve surgery, atrial septal closure and rhythm surgery), were used as

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Table 1: Baseline characteristics

	Total cohort			Matched cohort		
	Sternotomy *N* = 1776	MIMVS *N* = 725	SMD	Sternotomy *N* = 718	MIMVS *N* = 718	SMD
Age, mean (± SD)	64.3 (± 11.7)	64.0 (± 11.7)	-0.025	63.4 (± 12.4)	63.8 (± 11.6)	0.036
BMI, median [IQR]	25.6 [23.2–28.0]	24.8 [22.9–27.1]	-0.166	25.3 [23.1–27.7]	24.8 [22.9–27.2]	-0.054
BSA, median [IQR]	1.93 [1.79–2.06]	1.92 [1.76–2.07]	-0.060	1.93 [1.78–2.05]	1.93 [1.77–2.07]	-0.005
Male sex, *n* (%)	999 (56.3)	413 (57.0)	-0.014	410 (57.1)	411 (57.2)	-0.003
Diabetes, *n* (%)	147 (8.3)	30 (4.1)	-0.208	26 (3.6)	30 (4.2)	0.028
Chronic lung disease, *n* (%)	187 (10.5)	33 (4.6)	-0.287	35 (4.9)	33 (4.6)	-0.013
Extracardiac arteriopathy, *n* (%)	54 (3.0)	9 (1.2)	-0.162	6 (0.8)	9 (1.3)	0.038
Active endocarditis, *n* (%)	31 (1.7)	1 (0.1)	-0.433	1 (0.1)	1 (0.1)	<0.001
Serum creatinine (>200 μm/l), *n* (%)	21 (1.2)	2 (0.3)	-0.173	2 (0.3)	2 (0.3)	<0.001
LV function, *n* (%)						
Good	1195 (67.3)	618 (85.2)	–	616 (85.8)	612 (85.2)	–
Moderate	550 (31.0)	95 (13.1)	-0.529	93 (13.0)	95 (13.2)	0.008
Poor	31 (1.7)	12 (1.7)	-0.007	9 (1.3)	11 (1.5)	0.022
Pulmonary artery pressure, *n* (%)						
Normal	1430 (80.5)	632 (87.2)	–	631 (87.9)	627 (87.3)	–
Moderately increased	238 (13.4)	54 (7.4)	-0.227	55 (7.7)	54 (7.5)	-0.005
Severely increased	108 (6.1)	39 (5.4)	-0.031	32 (4.5)	37 (5.2)	0.031
EuroSCORE I, median [IQR]	3.61 [2.08–6.22]	2.99 [1.90–5.36]	-0.202	3.05 [2.03–5.16]	2.97 [1.89–5.22]	0.026
Tricuspid valve surgery, *n* (%)	495 (27.9)	80 (11.0)	-0.537	78 (10.9)	80 (11.1)	-0.008
Atrial septal closure, *n* (%)	49 (2.8)	23 (3.2)	0.024	23 (3.2)	22 (3.1)	0.009
Rhythm surgery, *n* (%)	373 (21.0)	100 (13.8)	-0.209	103 (14.4)	100 (13.9)	-0.012

BMI: body mass index; BSA: body surface area; EuroSCORE I: European System for Cardiac Operative Risk Evaluation; IQR: interquartile range; LV: left ventricular; MIMVS: minimally invasive mitral valve surgery; SD: standard deviation; SMD: standardized mean difference.

covariates for the propensity score matching model. A total of 718 pairs were matched in a 1:1 ratio. After matching, the 2 groups were comparable for all confounders (standardized mean difference <0.10).

Operative characteristics

Of the overall cohort ($n = 2501$), 1971 patients underwent MVr and 530 patients underwent MVR, resulting in a repair rate of 78.8%. Repair rate for the unmatched MST group was 80.0% vs 75.9% in the MIMVS group ($P = 0.02$). As the aetiology of mitral valve disease was not registered, this could not be related to the repair rate for either group. Isolated MV surgery was performed in 1598 patients (63.9%). For the matched cohorts, in the MST group, MVr was performed in 581 patients (80.9%) vs 548 patients (76.3%) in the MIMVS group ($P = 0.04$). Concomitant procedures are presented in Table 2. Unfortunately, CPB- and X-time exhibited excessive missing values (32.7% and 33.1%, respectively) and could not be included in the analysis. Operative characteristics are depicted in Table 2.

Short-term outcomes

In the overall cohort ($n = 2501$), 30-day mortality was 1.3% ($n = 33$) and stroke occurred in 1.3% of patients ($n = 33$). For the matched cohorts, 30-day mortality was 1.1% in the MST group, compared to 0.7% in the MIMVS group ($P = 0.58$). Stroke occurred in 1.1% of patients ($n = 8$) in the MST group, compared to 0.6% of patients ($n = 4$) in the MIMVS group ($P = 0.25$). Mortality and complication rates are depicted in Table 3. One-year mortality was 2.6% in the MST group compared to 2.1% in the MIMVS group ($P = 0.60$). There were no differences in complications as periprocedural myocardial infarction ($P = 0.48$), pneumonia ($P = 0.99$), renal failure ($P = 0.20$) or vascular complications ($P = 0.32$). A significantly reduced incidence of postoperative arrhythmia was observed in the MIMVS group (22.4% vs 32.3%, $P < 0.001$) compared to the MST group. Interestingly, length of hospital stay (LOHS) was shorter in the MST group (6 days [4–8] vs 6 days [5–8], $P = 0.001$).

Long-term outcomes

Survival. For the overall cohort, the median follow-up was 36 months [20–56], with a 5-year survival rate of 93.6%. For the matched groups, the survival rate at 5 years was 95.0% in the MST group, compared to 94.3% in the MIMVS group ($P = 0.49$, Fig. 1A).

Freedom from reintervention. In the overall cohort, 57 patients were reoperated during long-term follow-up, resulting in a freedom from MV reintervention of 97.7% at 5 years. During the initial procedure, these 57 patients underwent MVR (4 patients, all in the MST group) and MVr (53 patients). Procedures performed during reintervention were re-repair in 17 patients, MVR in 38 patients and a transcatheter procedure in 2 patients.

In the MST group, freedom from reintervention was 98.1% at 5 years (97.9% for the matched group), while this was 96.8% in the MIMVS group ($P = 0.001$, Fig. 2A). In the sub-group analysis, a significant difference also found in MV reintervention at 5 years when specified to actual isolated MVr (97.4% vs 95.8%, respectively, $P = 0.003$, Fig. 2A).

Reintervention-free survival. For the overall cohort, the median reintervention-free survival was 25 months [9–47], with a 5-year reintervention-free rate of 93.2%. For the matched groups,

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Table 2: Procedural characteristics

Total cohort	Sternotomy *N* = 1776	MIMVS *N* = 725	*P*-value	Sternotomy *N* = 718	MIMVS *N* = 718	*P*-value
Type of procedure, *n* (%)						
Mitral valve reconstruction	1421 (80.0)	550 (75.9)	**0.02**	581 (80.9)	548 (76.3)	**0.04**
Mitral valve replacement	355 (20.0)	175 (24.1)		137 (18.1)	170 (23.7)	
Tricuspid valve reconstruction	492 (27.7)	78 (10.8)	**<0.001**	75 (10.4)	78 (10.9)	0.80
Tricuspid valve replacement	3 (0.2)	2 (0.3)	0.59	3 (0.4)	2 (0.3)	0.99
Atrial septal closure	49 (2.8)	23 (3.2)	0.58	23 (3.2)	22 (3.1)	0.88
Rhythm surgery	373 (21.0)	100 (13.8)	**<0.001**	103 (14.3)	100 (13.9)	
Type of prothesis mitral valve, *n* (%)						
Mechanical valve	173 (48.7)	67 (38.3)	**0.02**	61 (44.5)	67 (39.4)	0.12
Biological valve	178 (50.9)	108 (61.7)		75 (54.7)	103 (60.6)	
Unknown	4 (1.1)	0 (0)		1 (0.1)	0 (0.0)	

MIMVS: minimally invasive mitral valve surgery.
Bold denotes statistical significance $p < 0.05$.

Table 3: Postoperative complications and follow-up

	Sternotomy *N* = 1776	MIMVS *N* = 725	*P*-value	Sternotomy *N* = 718	MIMVS *N* = 718	*P*-value
30-Day mortality, *n* (%)	28 (1.6)	5 (0.7)	0.08	8 (1.1)	5 (0.7)	0.58
1-year mortality, *n* (%)	60 (3.4)	15 (2.1)	0.08	19 (2.6)	15 (2.1)	0.60
Overall mortality, *n* (%)	119 (6.7)	42 (5.8)	0.40	36 (5.0)	41 (5.7)	0.64
Follow-up in days, median [IQR]	1135 [491–1706]	427 [239–1093]		1262 [656–1738]	951 [676–1661]	
Hospital stay in days, median [IQR]	5 [4–8]	6 [5–8]	**0.001**	6 [4–8]	6 [5–8]	**0.001**
Perioperative myocardial infarction, *n* (%)	10 (0.6)	3 (0.4)	0.77	5 (0.7)	3 (0.4)	0.48
Pneumonia, *n* (%)	42 (2.4)	15 (2.1)	0.65	16 (2.2)	15 (2.1)	0.99
Reintubation due to respiratory insufficency, *n* (%)	14 (0.8)	3 (0.4)	0.42	6 (0.8)	3 (0.4)	0.32
Prolonged intubation (>24 h), *n* (%)	37 (2.1)	6 (0.8)	**0.03**	18 (2.5)	6 (0.8)	**0.02**
Re-admission to ICU, *n* (%)	32 (1.8)	8 (1.1)	0.21	9 (1.3)	8 (1.1)	0.99
Cerebrovascular accident, *n* (%)	29 (1.6)	4 (0.6)	**0.03**	8 (1.1)	4 (0.6)	0.25
Kidney failure, *n* (%)	27 (1.5)	3 (0.4)	**0.02**	7 (1.0)	3 (0.4)	0.20
Gastro-intestinal complications, *n* (%)	10 (0.6)	4 (0.6)	0.99	4 (0.6)	4 (0.6)	0.99
Vascular complications, *n* (%)	3 (0.2)	1 (0.1)	0.99	3 (0.4)	1 (0.1)	0.32
New-onset arrhythmia, *n* (%)	576 (32.4)	162 (22.3)	**<0.001**	225 (31.3)	161 (22.4)	**<0.001**
Reexploration (within 30 days), *n* (%)	93 (5.2)	39 (5.4)	0.89	38 (5.3)	39 (5.4)	0.99
Reintervention, *n* (%)	34 (1.9)	23 (3.2)	0.06	15 (2.1)	23 (3.2)	0.25

ICU: intensive care unit; IQR: interquartile range; MIMVS: minimally invasive mitral valve surgery.
Bold denotes statistical significance $p < 0.05$.

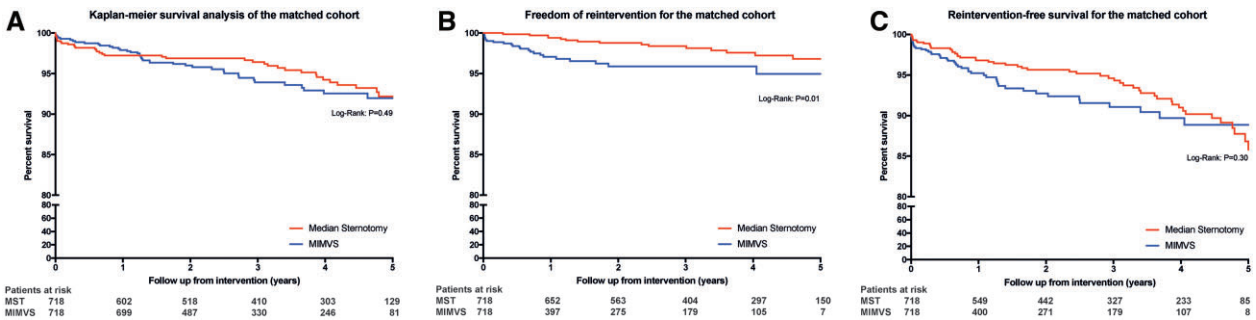

Figure 1: (**A**) Kaplan–Meier survival analysis of the matched cohort. (**B**) Freedom of reintervention for the matched cohort. (**C**) Reintervention-free survival for the matched cohort.

the reintervention-free survival rate at 5 years was 92.8% in the MST group, compared to 93.6% in the MIMVS group ($P = 0.30$, Fig. 1C).

In the sub-group analysis, when specified to actual isolated MVr, a reintervention-free survival rate of 93.1% vs 93.8% for MST and MIMVS was found, respectively ($P = 0.23$, Fig. 2B).

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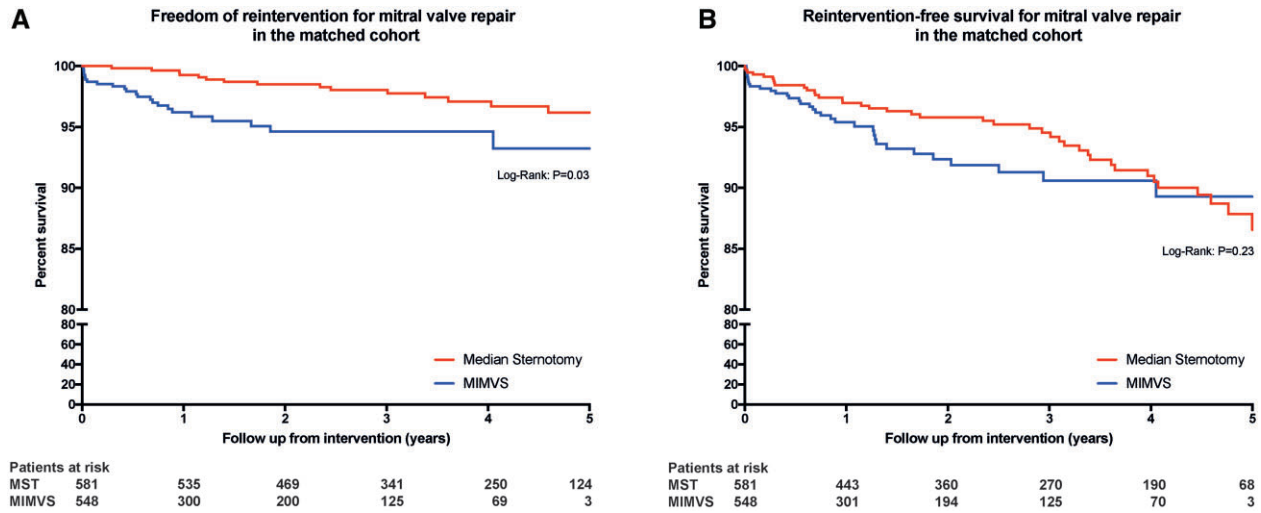

Figure 2: (**A**) Freedom of reintervention for mitral valve repair in the matched cohort. (**B**) Reintervention-free survival for mitral valve repair in the matched cohort.

DISCUSSION

This study is the first nationwide multicentred study to investigate the short- and long-term outcomes of MIMVS compared to MV surgery through MST. The most important finding is that MIMVS is as safe as MST in terms of short- and long-term outcomes, based on real-world data.

As is evident by the early results, isolated MV surgery, in all comers with a variety of mitral valve pathologies, is a safe procedure with a 30-day mortality rate of 1.3% for the total cohort (MIMVS and MST). This mortality rate is slightly lower, but in line with previous findings of the Society of Thoracic Surgeons database (2.9% early mortality) [23] and the Netherlands Heart Registration [24]. The difference can be explained by the different era during which these studies were conducted, as early mortality for mitral procedures has improved during the past decades [25], most probably due to an increased penetration of MVr [17]. When comparing an MIMVS approach to MST, we did not find a difference in short-term mortality (0.7% vs 1.1% in the matched cohorts, respectively), demonstrating the safety of MIMVS. Several large single-arm cohort and comparative studies (of which 1 was randomized [26]) have found similar results [8], which were confirmed in meta-analyses [3, 4]. In addition, the results of the current Dutch multicentre study are in line with a previous propensity matched multicentre cohort from the UK [10].

For all cardiac surgical procedures, stroke is one of the most feared complications. In earlier comparative studies, this complication seemed to occur more in patients undergoing MIMVS, compared to MST [27], potentially related to the use of retrograde perfusion [28]. With the advent of routine computed tomography scans for the screening of peripheral arterial disease [29], patient selection has improved and the occurrence of stroke as a complication has been diminished [30], irrespective of aortic occlusion techniques [31]. As such, the results of the current study are a reflection of contemporarily used techniques, as stroke rate was low and did not differ between both groups.

Interestingly, there was a statistically significant difference in LOHS between the MST and MIMVS groups (6 vs 6 days, respectively, due to distribution $P = 0.001$). As such, the difference does not seem to be clinically important. Still, a perceived advantage of MIMVS is the faster recovery time, in terms of ventilation duration, intensive care unit stay and LOHS [4]. However, the LOHS found in our study is still comparable to other high-volume centre experiences. As the current study is based on a multicentre registry, some inter-hospital differences in discharge policies might be expected which could be explanatory for these results. Furthermore, in the minimally invasive group, a larger proportion of bio prosthesis was implanted. These patients had a median age of 74 years [68–78] for median sternotomy and a median age of 77 years [72–81] for MIMVS.

Long-term follow-up of patients surgically treated for mitral disease, especially in the setting of MIMVS, can be challenging, as most of these procedures are performed in tertiary referral institutions. Subsequently, only single-centre experiences have reported long-term survival for patients undergoing MIMVS [1, 2]. Therefore, the current study is the first to evaluate and compare the long-term outcomes of MIMVS and MST approaches in the Netherlands. Of note, 5-year survival for the whole cohort was excellent in the Netherlands (93.6%), similar to previous reports [2], and long-term survival did not differ between the matched cohorts, also in line with Grant *et al.* [10], reporting British results.

In addition to safety, efficacy is as important to confirm the validity of MIMVS as a suitable alternative to MST. In this context, efficacy can be defined as the durability of the operation, which is often presented in literature by repair rate and freedom from mitral reinterventions.

A significant difference in repair rate was found for both the unmatched and matched population (80.0% vs 75.9% and 80.9% vs 76.3%) in favour of MST. The repair rate could unfortunately not be specified for mitral valve pathology. The difference could be a reflection of surgical experience and surgical volume, which is a known factor in the literature. Furthermore, in the analysed cohorts, mitral reintervention at long-term follow-up was increased in the MIMVS group (3.2% vs 2.1%, absolute difference of 8 patients). This difference did not disappear when correcting for isolated mitral repair. Perhaps, reintervention rate might not be an ideal marker of recurrence of mitral disease, as the acceptance for a reintervention potentially only reflects a lower surgical risk profile of the accepted patients. Ideally, recurrence of mitral

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regurgitation, its aetiology and its grading would be required to analyse actual differences between-group differences in depth.

Although the exact mechanism of recurrence of MV disease cannot be retrieved in the current registry, the observed difference should not be ignored. It may point to the difficulties accompanying the initiation of an MIMVS programme. MIMVS warrants sufficient case volume, as MVr and MIMVS have steep learning curves (75–125 cases) and require a steady patient base ensuring the continuity of 2 MIMVS procedures per week [16, 32]. As short- and long-term outcomes are dependent of operator and hospital volume [17, 19], these findings should stimulate a trend towards centralization of MIMVS procedures, performed by dedicated MV surgeons [33].

Therefore, we do not promote MIMVS as the preferred approach for all patients undergoing surgery for MV disease. Rather, patient selection for such an approach should be conducted through a standardized process of preoperative planning, minimalizing perioperative risks. Furthermore, we advocate for MIMVS to be performed by dedicated surgeons, optimizing durability and long-term survival.

Limitations

Although the current study design allows for a multicentre evaluation of all comers undergoing MV surgery, some remarks are appropriate. The NHR database was designed for quality evaluation, rather than for research purposes. As such, some important MV disease characteristics were not reported, such as MV disease aetiology, complexity and gradation. In addition, some variables, such as CPB- and X-time, were missing in a relatively large proportion as well. Furthermore, for patients undergoing MVr, no details regarding the repair technique (resection, neochordae or papillary muscle reposition) are registered, which is due to the nature of the registry. The techniques used could have influence on the long-term durability. As the registry was anonymized, and data could not be traced back to surgeons, the influence of surgeon volume could not be assessed. Finally, repair rate for degenerative MR influences short- and long-term outcomes, but its effect could not be evaluated in the current design.

CONCLUSION

The current study is the first nationwide registry to compare the survival of an MIMVS approach to MST for the treatment of MV disease in all comers. No difference in short-term mortality, long-term overall survival or reintervention-free survival was found. MIMVS is associated with a reduced risk of postoperative arrhythmia and prolonged intubation. An increased LOHS, a lower repair rate and an increased risk of reintervention over 5 years were observed for MIMVS. Patient selection, surgical volume and surgical expertise remain crucial.

Conflict of interest: Peyman Sardari Nia has a consultancy agreement with Neochord Inc, Edwards Lifesciences, Fuijfilm Medical and is the inventor of a mitral valve simulator that is commercialized through a start-up (Ma-trac) of Maastricht University Medical Center+. The other authors report no conflict of interest.

Data availability

All data were extracted for the database of the Netherlands Heart Registration. All data are available and accessible by the Cardiothoracic Surgery Registration Committee of the Netherlands Heart Registration.

Author contributions

Jules R. Olsthoorn: Conceptualization; Data curation; Formal analysis; Investigation; Methodology; Validation; Visualization; Writing—original draft; Writing—review & editing. **Samuel Heuts:** Data curation; Formal analysis; Methodology; Writing—original draft; Writing—review & editing. **Saskia Houterman:** Data curation; Formal analysis; Methodology; Writing—original draft; Writing—review & editing. **Jos G. Maessen:** Writing—original draft; Writing—review & editing. **Peyman Sardari Nia:** Conceptualization; Data curation; Formal analysis; Investigation; Methodology; Supervision; Validation; Visualization; Writing—original draft; Writing—review & editing.

Reviewer information

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REFERENCES

[1] Davierwala PM, Seeburger J, Pfannmueller B, Garbade J, Misfeld M, Borger MA *et al.* Minimally invasive mitral valve surgery: "The Leipzig experience". Ann Cardiothorac Surg 2013;2:744–50.
[2] Glauber M, Miceli A, Canarutto D, Lio A, Murzi M, Gilmanov D *et al.* Early and long-term outcomes of minimally invasive mitral valve surgery through right minithoracotomy: a 10-year experience in 1604 patients. J Cardiothorac Surg 2015;10:181.
[3] Sundermann SH, Sromicki J, Rodriguez Cetina Biefer H, Seifert B, Holubec T, Falk V *et al.* Mitral valve surgery: right lateral minithoracotomy or sternotomy? A systematic review and meta-analysis. J Thorac Cardiovasc Surg 2014;148:1989–95. e4.
[4] Modi P, Hassan A, Chitwood WR Jr. Minimally invasive mitral valve surgery: a systematic review and meta-analysis. Eur J Cardiothorac Surg 2008;34:943–52.
[5] Iribarne A, Easterwood R, Russo MJ, Wang YC, Yang J, Hong KN *et al.* A minimally invasive approach is more cost-effective than a traditional sternotomy approach for mitral valve surgery. J Thorac Cardiovasc Surg 2011;142:1507–14.
[6] Daemen JHT, Heuts S, Olsthoorn JR, Maessen JG, Sardari Nia P. Right minithoracotomy versus median sternotomy for reoperative mitral valve surgery: a systematic review and meta-analysis of observational studies. Eur J Cardiothorac Surg 2018.
[7] Galloway AC, Schwartz CF, Ribakove GH, Crooke GA, Gogoladze G, Ursomanno P *et al.* A decade of minimally invasive mitral repair: long-term outcomes. Ann Thorac Surg 2009;88:1180–4.
[8] McClure RS, Athanasopoulos LV, McGurk S, Davidson MJ, Couper GS, Cohn LH. One thousand minimally invasive mitral valve operations: early outcomes, late outcomes, and echocardiographic follow-up. J Thorac Cardiovasc Surg 2013;145:1199–206.
[9] Lange R, Voss B, Kehl V, Mazzitelli D, Tassani-Prell P, Günther T. Right minithoracotomy versus full sternotomy for mitral valve repair: a propensity matched comparison. Ann Thorac Surg 2017;103:573–9.
[10] Grant SW, Hickey GL, Modi P, Hunter S, Akowuah E, Zacharias J. Propensity-matched analysis of minimally invasive approach versus sternotomy for mitral valve surgery. Heart 2019;105:783–9.
[11] Paparella D, Fattouch K, Moscarelli M, Santarpino G, Nasso G, Guida P *et al.* Current trends in mitral valve surgery: a multicenter national comparison between full-sternotomy and minimally-invasive approach. Int J Cardiol 2020;306:147–51.
[12] Baumgartner H, Falk V, Bax JJ, De Bonis M, Hamm C, Holm PJ *et al.*; ESC Scientific Document Group. 2017 ESC/EACTS Guidelines for the management of valvular heart disease. Eur Heart J 2017;38:2739–91.
[13] Falk V, Cheng DC, Martin J, Diegeler A, Folliguet TA, Nifong LW *et al.* Minimally invasive versus open mitral valve surgery: a consensus

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statement of the international society of minimally invasive coronary surgery (ISMICS) 2010. Innovations (Phila) 2011;6:66–76.
[14] Ailawadi G, Agnihotri AK, Mehall JR, Wolfe JA, Hummel BW, Fayers TM *et al.* Minimally invasive mitral valve surgery I: patient selection, evaluation, and planning. Innovations (Phila) 2016;11:243–50.
[15] Wolfe JA, Malaisrie SC, Farivar RS, Khan JH, Hargrove WC, Moront MG *et al.* Minimally invasive mitral valve surgery II: surgical technique and postoperative management. Innovations (Phila) 2016;11:251–9.
[16] Holzhey DM, Seeburger J, Misfeld M, Borger MA, Mohr FW. Learning minimally invasive mitral valve surgery: a cumulative sum sequential probability analysis of 3895 operations from a single high-volume center. Circulation 2013;128:483–91.
[17] Bolling SF, Li S, O'Brien SM, Brennan JM, Prager RL, Gammie JS. Predictors of mitral valve repair: clinical and surgeon factors. Ann Thorac Surg 2010;90:1904–11, discussion 12.
[18] Gammie JS, Chikwe J, Badhwar V, Thibault DP, Vemulapalli S, Thourani VH *et al.* Isolated mitral valve surgery: the Society of Thoracic Surgeons adult cardiac surgery database analysis. Ann Thorac Surg 2018;106: 716–27.
[19] Chikwe J, Toyoda N, Anyanwu AC, Itagaki S, Egorova NN, Boateng P *et al.* Relation of mitral valve surgery volume to repair rate, durability, and survival. J Am Coll Cardiol 2017;69:2397–406.
[20] van Veghel D, Marteijn M, de Mol B, Measurably Better Study G, Advisory B; Measurably Better Study Group (The Netherlands) and Advisory Board. First results of a national initiative to enable quality improvement of cardiovascular care by transparently reporting on patient-relevant outcomes. Eur J Cardiothorac Surg 2016;49:1660–9.
[21] Fraser G, Yan R. Guided multiple imputation of missing data: using a subsample to strengthen the missing-at-random assumption. Epidemiology 2007;18:246–52.
[22] Benedetto U, Head SJ, Angelini GD, Blackstone EH. Statistical primer: propensity score matching and its alternatives. Eur J Cardiothorac Surg 2018;53:1112–7.
[23] Badhwar V, Rankin JS, He X, Jacobs JP, Gammie JS, Furnary AP *et al.* The Society of Thoracic Surgeons mitral repair/replacement composite score: a report of the Society of Thoracic Surgeons Quality Measurement Task Force. Ann Thorac Surg 2016;101:2265–71.
[24] Mokhles MM, Siregar S, Versteegh MI, Noyez L, van Putte B, Vonk AB *et al.* Male-female differences and survival in patients undergoing isolated mitral valve surgery: a nationwide cohort study in the Netherlands. Eur J Cardiothorac Surg 2016;50:482–7.
[25] Birkmeyer NJ, Marrin CA, Morton JR, Leavitt BJ, Lahey SJ, Charlesworth DC *et al.* Decreasing mortality for aortic and mitral valve surgery in Northern New England. Northern New England Cardiovascular Disease Study Group. Ann Thorac Surg 2000;70:432–7.
[26] Dogan S, Aybek T, Risteski PS, Detho F, Rapp A, Wimmer-Greinecker G *et al.* Minimally invasive port access versus conventional mitral valve surgery: prospective randomized study. Ann Thorac Surg 2005;79:492–8.
[27] Gammie JS, Zhao Y, Peterson ED, O'Brien SM, Rankin JS, Griffith B. J. Maxwell Chamberlain Memorial Paper for adult cardiac surgery. Less-invasive mitral valve operations: trends and outcomes from the Society of Thoracic Surgeons Adult Cardiac Surgery Database. Ann Thorac Surg 2010;90:1401–8.10e1, discussion 08–10.
[28] Murzi M, Cerillo AG, Miceli A, Bevilacqua S, Kallushi E, Farneti P *et al.* Antegrade and retrograde arterial perfusion strategy in minimally invasive mitral-valve surgery: a propensity score analysis on 1280 patients. Eur J Cardiothorac Surg 2013;43:e167-72–e172.
[29] Heuts S, Maessen JG, Sardari Nia P. Preoperative planning of left-sided valve surgery with 3D computed tomography reconstruction models: sternotomy or a minimally invasive approach? Interact CardioVasc Thorac Surg 2016;22:587–93.
[30] Leonard JR, Henry M, Rahouma M, Khan FM, Wingo M, Hameed I *et al.* Systematic preoperative CT scan is associated with reduced risk of stroke in minimally invasive mitral valve surgery: a meta-analysis. Int J Cardiol 2019;278:300–6.
[31] Bentala M, Heuts S, Vos R, Maessen J, Scohy TV, Gerritse BM *et al.* Comparing the endo-aortic balloon and the external aortic clamp in minimally invasive mitral valve surgery. Interact CardioVasc Thorac Surg 2015;21:359–65.
[32] Li J, Zhao Y, Zhou T, Zhu K, Zhai J, Sun Y *et al.* Learning curve of mitral valve repair: cumulative sum failure analysis from single high-volume center. J Thorac Dis 2020;12:6563–72.
[33] Heuts S, Olsthoorn JR, Hermans SMM, Streukens SAF, Vainer J, Cheriex EC *et al.* Multidisciplinary decision-making in mitral valve disease: the mitral valve heart team. Neth Heart J 2019;27:176–84.

APPENDIX

The members of the Cardiothoracic Surgery Registration Committee of the Netherlands Heart Registration are as follows:

Dr. S. Bramer, Amphia
Dr. W.J.P. van Boven, Amsterdam UMC, locatie AMC
Dr. A.B.A. Vonk, Amsterdam UMC, locatie VUmc
Dr. B.M.J.A. Koene, Catharina Ziekenhuis
Dr. J.A. Bekkers, Erasmus MC
Dr. G.J.F. Hoohenkerk, HagaZiekenhuis
Dr. A.L.P. Markou, Isala
Dr. A. de Weger, Leids Universitair Medisch Centrum
Dr. P. Segers, Maastricht UMC+
Dr. F. Porta, Medisch Centrum Leeuwarden
Dr. R.G.H. Speekenbrink, Medisch Spectrum Twente
Dr. W. Stooker, OLVG
Dr. W.W.L. Li, Radboudumc
Dr. E.J. Daeter, St. Antonius, Ziekenhuis
Dr. N.P. van der Kaaij, UMC Utrecht
Dr. G. Vigano, Universitair Medisch Centrum Groningen

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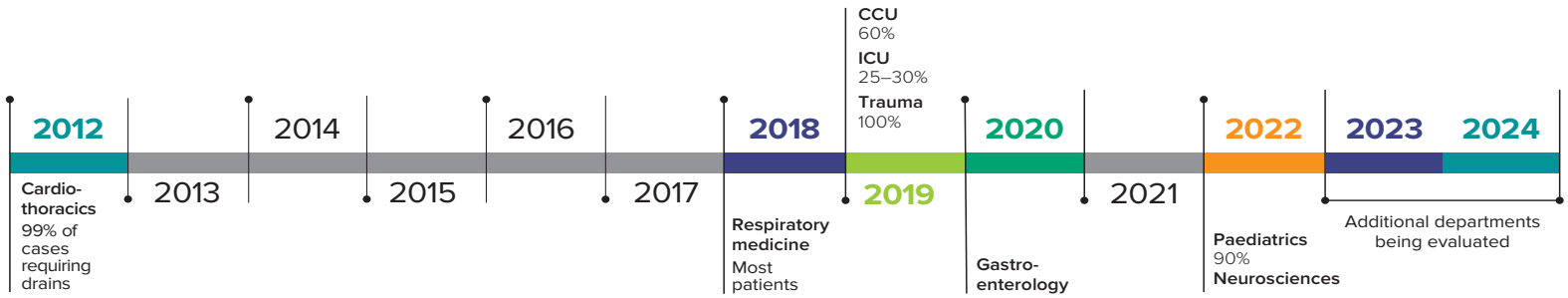
Real-world experience with Thopaz+

The Oxford University Hospitals NHS Foundation Trust experience

Thopaz+ is a portable digital chest drainage and monitoring system developed by Medela. It offers continuous objective monitoring of fluid loss and air leaks, which facilitates assessment of patients' progress, as well as standardisation of chest drainage management across different departments.[1] Clinical evidence has demonstrated that Thopaz+ is a useful tool in the management of patients that require chest drains and has clear clinical advantages compared with underwater seal drains.[1–3]

Thopaz+ and its predecessor, Thopaz, have been used within the Cardiothoracic Department at Oxford University Hospital NHS Trust since 2012. A report on this experience contributed to National Institute for Health and Care Excellence (NICE) Medical Technology Guidance 37.[1,4] Use of Thopaz+ in Oxford has since expanded to other departments within the trust. This document summarises the experience with Thopaz+ based on interviews with healthcare professionals (HCPs) at Oxford University Hospital NHS Trust in February/March 2024.

*Evolution of Thopaz+ use in Oxford: initial introduction by department and current usage**



*Percentage of cases using Thopaz+, where known from interviews.

CHEST DRAINAGE PROTOCOLS

Each department has a chest drain protocol based on their use of Thopaz+ or underwater seal drains, and whether active suction or physio mode is needed.

MOBILISATION

Improved and earlier mobilisation is a major advantage of Thopaz+ in relation to complications associated with immobility.

OBJECTIVE AND CONTINUOUS MONITORING LEADS TO IMPROVED DECISION-MAKING

Continuous monitoring improves chest drain decision-making by providing objective estimates/measurement of leakage. It helps determine when air leaks are resolving (allowing for earlier drain removal and discharge planning) or when further intervention is needed (such as referral to a surgeon).

LENGTH OF STAY

Digital drainage facilitates day-case procedures by giving HCPs confidence that their patients have no persistent air leaks or fluid loss.

RESPIRATORY

70% of patients following pleural intervention and 60% undergoing thoracoscopy return home the same day.

CORONARY CARE UNIT (CCU)

Length of stay of 7 days with Thopaz+ compared with 10 days with underwater seal drains.

THROUGHOUT THE PATIENT JOURNEY

Thopaz+ can be used throughout a patient's journey, which can reduce the possibility of issues and errors, because drains can become kinked or displaced whenever a device is changed. Suction can be added to a Thopaz+ device set up to provide straightforward drainage simply by pressing a button to initiate suction via the device itself.

COSTS AND EFFICIENCIES

The use of the device can lead to improved operational efficiencies and cost savings, which may justify the acquisition costs. From an evidence-based practice project in the USA, a digital air leak detection device after pulmonary lobectomy led to cost savings of $2,659 per hospital day.[5]

IMPROVED PATENT SAFETY

Thopaz+ is a closed system, reducing incidents, errors, mishaps, and infections. As a dry system, Thopaz+ prevents issues with water and device positioning. Non-medical staff can manage Thopaz+ if it is knocked over, with no patient impact. Thopaz+ has its own suction source, preventing complications with wall suction becoming displaced or unclipped.

STAFF EXPERIENCE

Precise fluid and air leak measurements including time trends, improve clinician confidence and decision-making and facilitate continuity of care. The user-friendly interface makes it easier to track air leaks and fluid output. Nursing time is saved with easy canister replacement, reduced manual monitoring, and visual and audible notifications alert HCPs of issues.

PATIENT EXPERIENCE

Patients can move around freely without nursing or healthcare assistant support. Earlier discharge reduces hospital stay. Patients can monitor their progress in terms of reducing volumes of fluid and air leaks on the display.

Summary of the real-world experience with Thopaz+

The experience of HCPs within Oxford University Hospitals NHS Foundation Trust over the past 12 years has shown that Thopaz+ has multiple benefits in the right circumstances and should be available for the vast majority of patients requiring a chest drain.

Francesco Di Chiara MD, MS THOR (Hons), FEBTS
Consultant Thoracic Surgeon Oxford University Hospitals NHS Foundation Trust

> Overall, our experience at Oxford University Hospitals NHS Foundation trust has shown that Thopaz+ is an indispensable asset for HCPs, redefining standards of care and operational efficiency across multiple medical departments. We encourage all units using chest drains to consider making the move from underwater seal drains to Thopaz+ in the vast majority of patients requiring chest drainage.

Quotes from interviews with a number of healthcare professionals at Oxford University Hospital NHS Trust:

> From the NHS perspective, I think it probably allows us to make earlier decisions about withdrawing chest drains and getting peopleout of hospital earlier.

> There are a number of ways to recoup the costs: efficiencies in the system, less litigation because things don't go wrong, staff sickness due to back injuries, and length of stay if you can get patients home quicker.

References

1. NICE. *Thopaz+ portable digital system for managing chest drains*. Medical technologies guidance 37. London: NICE, 2018 (updated 2022). (assessed March 2025)
2. Abdul Khader A *et al*. *J Thorac Dis* 2023;**15**:3776–82.
3. Frediani S, Romano G, Pardi V *et al*. *Front Pediatr* 2023;**11**:1280834.
4. Mitchell J. *Adopting Thopaz+ portable digital system for managing chest drains on the cardiothoracic ward at Oxford University Hospitals NHS Foundation Trust*. (assessed March 2025)
5. Patel C *et al*. Use of a digital air leak detection device to decrease chest tube duration. *Critical Care Nurs* 2023;**43**:11–21.

Read the full report:

The summary report has been written by HSJ Advisory on behalf of Medela AG, reflecting the views expressed in interviews with healthcare professionals. Medela AG funded the project and had input into the development of this report.

HSJ Advisory